

Quiz 1: DIC on Discrete Structures

20 marks, 60 min

28 Aug 2025

1. (5 marks) True or false. If true, give a short proof, if false give a counter-example and a short justification why its a counter-example.

(a) For all $a, b, c \in \mathbb{Z}$, if $(a \times b) + c^2$ is odd, then at least one of a or b or c must be odd.

Solution

True. **0.5 mark**

We shall prove the contrapositive of this, i.e. if all a, b, c are even, then $(a \times b) + c^2$ must be even. Putting $a = 2a'$, $b = 2b'$, $c = 2c'$, we get $(2a' \times 2b') + (2c')^2 = 2(2a'b' + 2c'^2)$, hence even, hence proved. **1.5 mark**

(b) For all sets A, B, C , it is the case that $A \cap (B \cup C) = A \cup (B \cap C)$

Solution

False. **0.5 mark**

Counterexample: $A = \phi$, $B = C = \{0\}$. **1 mark**

Note that counter-examples are compulsory. If correct Venn diagrams are drawn but no explicit example is given, 0.5 mark has been reduced

(c) There exist three distinct infinite sets A, B, C such that $A \cap (B \cup C) = A \cup (B \cap C)$.

Solution

True. **0.5 mark**

Example: $A = \mathbb{N} \setminus \{0, 1\}$, $B = \mathbb{N} \setminus \{0\}$, $C = \mathbb{N} \setminus \{1\}$. **1 mark**

If some sufficient property of sets for a valid example is given, but the existence of such sets is not proved, 0.5 mark deducted.

2. (3 marks) Prove the following by using induction, strong induction or WOP. If you use strong induction, write clearly where/why you need it.

Define a sequence $a_1 = 3, a_2 = 18, a_n = 6a_{n-1} - 9a_{n-2}$ for all $n \geq 3$.

Prove that 3^n divides a_n for all $n \geq 1, n \in \mathbb{N}$.

Solution

(a) **Proof By Induction** We prove by strong induction on n that $3^n \mid a_n$ for all $n \geq 1$.

Base cases. For $n = 1$: $a_1 = 3$, so $3^1 \mid a_1$. For $n = 2$: $a_2 = 18$, so $3^2 \mid a_2$.

Base cases - 1 mark (0.5 for each of the base cases, $n = 1$, and $n = 2$)

Inductive step. Assume as induction hypothesis that for some $N \geq 3$ we have $3^m \mid a_m$ for every $m < N$. In particular,

$$3^{N-1} \mid a_{N-1} \quad \text{and} \quad 3^{N-2} \mid a_{N-2}.$$

1 mark for correct induction hypothesis, if assumed just for a single m , then no marks

Write $a_{N-1} = 3^{N-1}k$ and $a_{N-2} = 3^{N-2}\ell$ for some integers k, ℓ . Using the recurrence,

$$a_N = 6a_{N-1} - 9a_{N-2} = 3(2a_{N-1} - 3a_{N-2}).$$

Substitute the factored forms:

$$\begin{aligned} 2a_{N-1} - 3a_{N-2} &= 2 \cdot 3^{N-1}k - 3 \cdot 3^{N-2}\ell \\ &= 3^{N-2}(2 \cdot 3k - 3\ell) \\ &= 3^{N-1}(2k - \ell). \end{aligned}$$

Hence

$$a_N = 3 \cdot 3^{N-1}(2k - \ell) = 3^N(2k - \ell),$$

since $2k - \ell \in \mathbb{Z}$, we have $3^N \mid a_N$.

Conclusion. By strong induction, 3^n divides a_n for every $n \geq 1$.

1 mark for the correct calculation

Other Common Mistakes and their grading policy.

Range of N in Induction Hypothesis : Any range other than $N \geq 3$ is invalid as it leaves some things unproven. **No/Wrong range gives 0.5 mark penalty.**

- if the range was anything with $N \geq 4$ but not $N = 3$ then the proof lacks a crucial part that $P(1) \wedge P(2) \rightarrow P(3)$
- if the range was anything that include $N = 1$ or $N = 2$ then one can't use the fact $a_N = 6a_{N-1} - 9a_{N-2}$, hence they need to be handled carefully. It is clear that you handle them as base-case but this still makes you Induction Hypothesis and Induction Step inconsistent.

Missing n=1, n=2 : Some people have started with base case $n = 3$, and then continued to prove. This approach is completely valid. But note that this will only conclude the statement is true for all $n \geq 3$ and doesn't talk about $n = 1, n = 2$, **So you are needed to explicitly write these 2 cases, Not writing them has 0.5 mark penalty.**

Using induction with wrong inductive step : If you prove a_N is true only from a_{N-1} then probably you went wrong somewhere in the proof. **marks are awarded for base case and induction hypothesis if they are consistent with the proof, and capped at a total of 1 mark (0.5 for base-case, 0.5 for Hypothesis)**

Not mentioning what k,l are : Missing to $k, \ell \in \mathbb{Z}$ is a mistake while writing formally, everything used should be correctly described, for this question it was obvious to assume this, but this is not true in a general setting. **No penalty.**

- (b) **Proof by Well-Ordering Principle** We are going to assume that the statement is false, and deduce a contradiction.

Let S be the set of natural numbers $i(\geq 1)$ for which 3^i doesn't divide a_i . We know that set S is non-empty by our assumption. Hence there exist a minimum element in S say k .

$$S = \{i \in \mathbb{N} : i \geq 1, 3^i \mid a_i\}$$

1 mark for correctly constructing the set and using Well-Ordering principle correctly. 0.5 mark penalty if you miss to say that S is non-empty.

We show that $k \geq 3$, as $i = 1, 2$ doesn't belong to the set,

$$\begin{aligned} a_1 = 3, 3^1 = 3 &\implies 3^1 \mid a_1 \implies 1 \notin S \\ a_2 = 18, 3^2 = 9 &\implies 3^2 \mid a_2 \implies 2 \notin S \end{aligned}$$

1 mark if correctly shown that $k \geq 3$

Since $k \geq 3$ we know by the definition that,

$$a_k = 6a_{k-1} - 9a_{k-2}$$

As k is minimum element of S , $k \geq 3$, we can conclude $1 \leq k-1, k-2 < k$ doesn't belong to S , Hence by definition of S , we have

$$3^{k-1} \mid a_{k-1} \quad \text{and} \quad 3^{k-2} \mid a_{k-2}.$$

Write $a_{k-1} = 3^{k-1}p$ and $a_{k-2} = 3^{k-2}q$ for some integers p, q . Using the recurrence,

$$a_k = 6a_{k-1} - 9a_{k-2}$$

Substitute the factored forms:

$$\begin{aligned} 6a_{k-1} - 9a_{k-2} &= 6 \cdot 3^{k-1}p - 9 \cdot 3^{k-2}q \\ a_k &= 3^k(2 \cdot p - q) \end{aligned}$$

We have proved that $3^k \mid a_k$, so $k \notin S$, But $k \in S$, Hence we have a contradiction. With this we can conclude that the only assumption of the problem statement being invalid is wrong. hence, 3^i divides $a_i \forall i \in \mathbb{N}$

1 mark for correct calculation.

Other Common Mistakes and their grading policy.

Wrong set S : Considering set S ,

$$S = \{a_i : i \in \mathbb{N}, i \geq 1, 3^i \mid a_i\}$$

Using this and then considering a_{k-1} and a_{k-2} not a part of the set requires that they are less than a_k proving which is again as hard as the question itself, this makes the entire proof worthless. **Similar errors in future will result to harsh penalty.**

0.5 mark penalty.

3. (7 = 4 + 1.5 + 1.5 marks) Let $f : A \rightarrow B$ be an injection and $g : B \rightarrow C$ a bijection.

(a) Prove or disprove: There must exist a surjection from C to A . Write all steps carefully.

Solution (with assumption)

Note that the following proof only works if you assume that A is non-empty. Under this assumption, here is the solution.

Let $f : A \rightarrow B$ be injective and $g : B \rightarrow C$ be bijective. Since g is bijective, its inverse $g^{-1} : C \rightarrow B$ exists.

Because f is injective, the restriction $f : A \rightarrow f(A)$ is a bijection, where $f(A) = \{b \in B \mid \exists a \in A, f(a) = b\}$. Hence it admits an inverse $f^{-1} : f(A) \rightarrow A$. Fix some $a_0 \in A$ (this is where we use the assumption that $A \neq \emptyset$) and define

$$s : B \rightarrow A, \quad s(b) = \begin{cases} f^{-1}(b), & b \in f(A), \\ a_0, & b \in B \setminus f(A). \end{cases}$$

(It is clear that s is surjective.)

2 marks: Constructing s

Now define

$$h = s \circ g^{-1} : C \rightarrow A.$$

This is well defined since s and g^{-1} are.

1 mark: Constructing h

For any $a \in A$, we have

$$h(g(f(a))) = s(f(a)) = a,$$

so every element of A is in the image of h . Therefore $h : C \rightarrow A$ is surjective.

1 mark: Proving h is surjective

We have deducted 0.5 marks from this question's total (given they are non-zero) if you omitted to mention regarding non-emptiness of A (see below for what will happen when A is empty). On the other hand, if you have explicitly mentioned this assumption, then we have given full marks.

Solution

If A is empty, then it is not possible to find such a surjective function (domain C is non-empty, and range is empty).

- (b) Will there always exist a bijection from C to A ? Why or why not?

Solution

Let

$$A = \{1, 2\}, \quad B = \{1, 2, 3\}, \quad C = \{a, b, c\}.$$

Define the injection $f : A \rightarrow B$ by $f(1) = 1$, $f(2) = 2$, and the bijection $g : B \rightarrow C$ by

$$g(1) = a, \quad g(2) = b, \quad g(3) = c.$$

Then f is injective and g is bijective, so the hypotheses holds. The composition in the proof above yields a surjection $h : C \rightarrow A$. However $|A| = 2$ and $|C| = 3$, so no bijection $C \rightarrow A$ can exist.

0.5 mark: for the correct answer (bijection does not exist)

1 mark for a correct counter example

No partial marks for this part.

- (c) If C is countably infinite, then prove that A must be countable.

Solution

We have two cases, either A can be finite or infinite.

Case 1: if A is finite, then it is always countable.

Case 2: If A is infinite, then there exists an injection h from $\mathbb{N}$ to A (as shown in class and as given in Instructions to Quiz).

But $f : A \rightarrow B$ is an injection and $g : B \rightarrow C$ is an injection (since it is a bijection it must be also an injection), hence $f \circ g : A \rightarrow C$ is an injection (since composition of injections is an injection).

Since, C is countably infinite, there exists a bijection $h' : C \rightarrow \mathbb{N}$, which is also an injection.

Thus $f \circ g \circ h'$ is an injection from A to $\mathbb{N}$.

Thus we have shown there exist injections from $\mathbb{N}$ to A and vice versa. So by Schroder-Bernstein Theorem, we conclude that there must exist a bijection from $A \rightarrow \mathbb{N}$, i.e., A is countably infinite and hence countable.

One may also proceed by saying that there exists a surjection from A to $\mathbb{N}$, and from prove that there must be a surjection from N to A , and conclude that A will be countable.

If the arguments are shown only on finite sets or via size/cardinality (and not functions), 0 marks are awarded.

If the case of A being finite has been omitted, and only arguments for infinite sets is shown, then 0.5 marks have been deducted.

4. (5 marks) Let $n \in \mathbb{N}$, $n \geq 1$. In any set of $n + 1$ positive integers, where each integer is at most $2n$, show using induction/WOP that there must exist two integers i, j , $i \neq j$ such that i divides j .

Solution

Solution using Induction

We prove by induction on n the statement

$P(n)$: In any set of $n + 1$ positive integers, each $\leq 2n$, there exist $i \neq j$ with $i \mid j$.

Base case. $n = 1$. Any set of 2 positive integers each ≤ 2 must contain $\{1, 2\}$, so one divides the other. Thus $P(1)$ holds.

Inductive step. Assume $P(n)$ is true for some $n \geq 1$. We prove $P(n + 1)$: Any set S of $n + 2$ positive integers each $\leq 2(n + 1) = 2n + 2$ contains two with one dividing the other.

Consider such a set S . There are two cases.

Case 1: At most one of $2n + 1, 2n + 2$ belongs to S . Removing that element leaves $n + 1$ numbers all $\leq 2n$. By the inductive hypothesis $P(n)$ these $n + 1$ numbers contain a pair with one dividing the other, so S does as well.

Case 2: Both $2n + 1$ and $2n + 2$ belong to S . Suppose, for a contradiction, that no two elements of S are in a divisibility relation. Then $n + 1 \notin S$, because $n + 1 \mid 2n + 2$. Now remove $2n + 1$ and $2n + 2$ from S and insert $n + 1$ instead; call the new set S' . Then S' has $n + 1$ elements, each $\leq 2n$.

We claim S' has no divisibility pair. The elements of S' consist of the n elements of S other than $2n + 1, 2n + 2$, together with $n + 1$. By assumption, the n retained elements do not divide one another. None of them divides $n + 1$, for if $x \mid (n + 1)$ then since $n + 1 \mid (2n + 2)$, we would have $x \mid (2n + 2)$, contradicting the assumption. Finally, $n + 1$ does not divide any retained element $y \leq 2n$, because the only multiple of $n + 1$ not exceeding $2n$ is $n + 1$ itself (and $n + 1 \notin S$ originally). Thus S' has no divisibility relation.

But S' is a set of $n + 1$ integers from $\{1, 2, \dots, 2n\}$, contradicting the inductive hypothesis $P(n)$.

Therefore our assumption was false, and S must contain two elements with one dividing the other.

By induction, the statement $P(n)$ holds for all $n \geq 1$.

Solution using WOP: Let $P(n)$ be the property that any set with $n + 1$ elements, all less than or equal to $2n$ has 2 numbers i, j where $i \neq j$ and $i|j$ and let S be the set of all natural numbers n except 0, where $P(n)$ does not hold.

Let us assume S is non-empty. Then according to WOP, S has a smallest element, call it k . For $n = 1$, the only set possible is $\{1, 2\}$ in which $1|2$ hence k can not be 1.

Since k can't be one, $k - 1 \geq 1$ and since k is the smallest element where $P(k)$ is false, $P(k - 1)$ is true.

Let X be some $k + 1$ length subset of the first $2k$ positive naturals for which no number divides any other.

Case 1: If X does not contain either of $2k$ and $2k - 1$, there will be at-least $k - 1$ numbers less than $2(k - 1)$ that all don't divide each other. But then those set of numbers would be a counterexample for $P(k - 1)$ to hold.

Case 2: If X contains both $2k$ and $2k - 1$, consider k and all of its divisors. None of these elements can lie in X , as otherwise that element would divide $2k$. But since k does not have any larger multiples of itself that are less than $2(k - 1)$, and none of its divisors are present, adding k to $X - \{2k - 1, 2k\}$ would give a set that is a counter-example to $P(k - 1)$.

Hence since k was the smallest element for which $P(k)$ is false, we arrive at a contradiction, and our assumption of S being non-empty was wrong. So S is empty, and $P(k)$ is true for all positive integers k .

- 0.5 mark for the correct base case (Even for WOP, it would be needed to show that the smallest element (x) is not the base case itself, otherwise we can't make any argument for numbers smaller than this as there won't be any smaller numbers!)

- 1 mark for the inductive step, no partials, if used WOP, then again 1 mark for the correct WOP statement

- 1.5 marks for the first case, no partials if the arguments do not make sense. Here $1.5 = 0.5 + 0.5 + 0.5$ where 0.5 is for when $n + 2$ elements are $\leq 2n$, 0.5 for when one is $2n + 1$ and rest $\leq 2n$ and 0.5 for when one is $2n + 2$ and rest $\leq 2n$

- 2 marks for the second case, no partials if the arguments do not make sense. $2 = 1 + 1$ where 1 is for building the correct S' and 1 is for showing how is S' linked to S

- If neither induction nor WOP was used, then a correct solution (using other techniques) was given between 2-3.5 max marks for stating all arguments correctly.

- **Common Mistake:** For the case 2 (as in the induction solution), many students tried to prove it by removing $2n + 1$, $2n + 2$ recursively and told that they would eventually reach the base case. But that is wrong. Take for e.g. $\{2, 3, 5, 6\} \rightarrow \{2, 3\}$.

Also for quite some solutions, students have tried by taking cases like 1 is there, 1 is not there, 2 is there, etc. But no solution properly explained the "2 is not there" case. And we have not provided partial marks just for writing trivial subcases like "1 is there/not there. 2 is there" unless "2 is not there" is proved as well.